BikeWatch: Anti-Theft Bike Alarm

User Guide

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Preface

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Audience

This user guide is meant for general information for those who in addition to the usual bike lock, would like extra security to prevent bike theft through a smarter anti-theft system like an alarm to deter potential thieves.

Key Terms

- Arduino Nano 33 BLE Sense: Arduino board used for the hardware alarm device.
- ARM® Cortex™-M4 32 bit processor: The processor that is on the nRF52840 chip
- Armed: state of device where if the bike is moved without consent from the owner, the device will trigger the alarm.
- BLE: Bluetooth Low Energy - conforms to protocol which allows it to communicate with smartphones.
- Disarmed: state of device where the bike can be moved without triggering the alarm.
- GPIO: General purpose input/output used to connect processor chips to the development board.
- LSM9DS1: 3 Axis Accelerometer Gyroscope Module that will be used to detect movement.
- Maxell CR2032 Lithium Batteries: Lithium batteries that will be able to have a long battery life and support the voltage requirement for our device.
- nRF52840 chip: MicroController that consists of the processor, BLE chip and protocol transceiver.
- Piezo Buzzer: Horn used to make the alarm sound when motion is detected.
- UUID: Universally Unique Identifier, a 128-bit identifier used to identify information in computer systems.

Product Overview

Background

In Davis, California, the bike riding capital of the United States, most people own and ride a bicycle in some capacity. Due to this, bicycle theft is a prevalent and costly issue for many riders. In order to keep bikes secure, we suggest that in addition to using a lock, we design a hidden bike alarm that goes off when the bike is in unforseen motion.

Description

Through this bike alarm, we attempt to decrease the chance of bicycle theft. The loud siren of the alarm would deter potential thieves by bringing attention towards the bike. The alarm would be controlled via a bluetooth connection through native Android and iOS mobile applications. However, once the bluetooth connection is disconnected, usually to distance between bike and owner, the bike becomes armed. The bike would then be disarmed when the bluetooth connection is re-established. In addition, once in proximity the owner could arm the alarm even if they are within the bluetooth's range. Additionally, the mobile apps are connected to a Firebase real time database in order to report users stolen bikes and check for stolen bikes in the area. The alarm can be hidden in a discrete location such as the handle bars, or in the shaft under the bike seat. Bike theft is an issue that no bike owner should have to deal with, and with our project we think we could decrease the rate of theft.

Technical Specifications

Database: Firebase
Using a real time database to easily read and write data to and from the mobile applications, perfect for our use since it updates the list of stolen bikes as they are reported and updated after they are found.
Hardware components: Arduino Nano 33 BLE Sense, Maxell CR2032, Piezo Buzzer
Using a nrf52840 core Bluetooth Low Energy chip with an ARM® Cortex™-M4 32 bit processor is perfect for the project with its on board motion and gesture sensors in the LSM9DS1 gyroscope and accelerometer for creating and checking for motion passwords as well as triggering the Piezo Buzzer, which is connected serially to the Arduino board to produce the alarm. Since the chip is also a BLE chip, it also fulfills the requirement of a long lasting device by requiring very little power. The chip is programmed using the Arduino IDE and libraries in order to create the proper behavior in the device. The Maxell CR2032 lithium battery is used since they give a long battery life and support the voltage requirement for the Nano 33 BLE Sense device, as well as allow it to be rechargeable.
Mobile iOS Application: Swift, Cocoapods
Swift is used to write the native iOS application and Cocoapods is used to interface the Firebase libraries and include the dependencies needed to read and write from the database.
Mobile Android Application: Java
Java is used for its ease of use and extensive documentation within Android development in order to write the native Android application.

Features

Background Process
To allow for constant connection while in range of the user's hardware device, the application can run in the background while not currently open on the user's phone.

Rechargeable Battery

To allow ease of access for the user, the hardware alarm device can be recharged since it is a lithium battery.

Bluetooth Connection Between Phone App and Bike Device

Users can connect their hardware alarm device to their phone application to perform various operations such as registering the bike, arming and disarming the device, swapping between bikes, as well as reporting stolen bikes.

Registering Bikes/Devices with QR Code

Using a supplied QR code that comes with the alarm device, the device's UUID is supplied to the mobile application so that it can be registered and interact via a Bluetooth connection.

Registering Bikes/Devices with UUID

The user can enter the UUID of the hardware alarm through a textbox to supply it to the mobile application in order to register the device and interact via a Bluetooth connection.

Registering Multiple Bikes

Users can have more than one bike or device registered. They must own separate hardware device alarms in order to do so.

Swapping Between Active Bikes

Through the menu item holding information on the current bike, the user can swap between the active bikes that they have registered through the application.

Arming & Disarming Alarm

The state of the hardware bike alarm can be updated through the mobile application and the motion password. The alarm can be toggled to be armed and unarmed using the phone application. Additionally, when the user and the device are no longer connected, if the user left the device unarmed, the device is toggled to armed.

Motion Based Password

In the event the user's mobile phone is out of battery or they do not want to take the phone out of their pocket, a motion based password made from 5 distinct movements (of which can be a left, right, up, and/or down movement of the bike) set by the owner of the bike to allow the user to disarm the hardware alarm in order to allow for easy riding without taking out the mobile phone.

Location of Bike

Through the application, the user can see the current location of their bike and hardware device as well as where they currently are. When the bike and user are disconnected from their Bluetooth connection, the user can see the location of the bike as the location where the two as a pin where they lost Bluetooth connection with the bike.

Device Battery Life

The user can see the percentage of time left on the battery of the hardware alarm through the mobile application. The mobile application receives data from the Bluetooth connection to tell the phone how much battery life is left in the device before it needs to be recharged. Below is how we calculated the battery life percentage:

Each input output takes 15mA and the device utilizes two output ports. So when the device is armed it is using 15mA and when the alarm is going off an addition 15mA is used. Most of the

time the alarm will not be sounding. The battery can last for 80 hours of constant use; so if the user was to arm the device for 3 hours a day, 5 days a week, the device's battery would last for 1 month without charge due to the fact that it holds 2600mA. Depending on how much the alarm goes off.

Reporting Stolen Bikes
In the event the user's bike is stolen, the user can send to the database a report of the user's bike that was stolen as well as a description of the bike.
Finding Stolen Bikes
Through a background process in the application, users can search for missing bikes reported in the application by searching for UUIDs within range of the user and their application provided to the Firebase real-time database through reporting of bikes by other users. The user who reported the stolen bike is notified of the location of their bike.

Project Components

Hardware alarm: The first component of our alarm system is a hardware device to detect movement and abnormal movements when armed. When in these unauthorized movements, the alarm emits a loud noise in order to bring attention in the direction of the bike and anyone tampering with the bicycle. This device is inserted within the frame of the bike and checks for motion. When connected via Bluetooth to our mobile application it can be armed and disarmed through the application as well as through a specific set of motions that are set by the user.
Mobile Application: Alongside the hardware device, there also is a mobile application in order to toggle the alarm on and off as well as to switch devices (bikes) currently in use if the user has multiple bikes. Additionally, the application also is able to register new alarms for their bikes through the application using a QR code or by listing out the devices UUID, as well as prompt the hardware alarm for different modes such as setting the motion password. In addition, the application runs a background process to search for bikes that have been reported as stolen to the database in order to notify the bike's user of its location when found by the application. The list of stolen bikes is read at regular intervals in order to have current information on the stolen bikes in the database.
Database: Used to store data such as the UUID and description of the bike that was self-reported by the owner as stolen through the application. This list of data is seen by other users' applications in order to crowdsource the effort to find the device and bike through the mobile application.

Installation and Distribution

Prerequisites

Requires a mobile phone running at the minimum Android OS 4.4 or iOS 13. The put together hardware alarm device is also required.

Basic Setup

First, the mobile application will need to be downloaded from the Google Play Store or the Apple App Store. Additionally, the device must be purchased in order for it to be used.

Registering a Bike

To register your bike alarm with the application, there is a QR code included as well as the UUID of the alarm.

QR Code

Using the QR code included, you can scan it within the app after choosing to register a bike within the app. This gives the UUID to the application and is stored within the apps data.

Using UUID

Using the UUID included, you can type it into a textbox in the app to register the UUID of the device included. This gives the UUID to the application and is stored within the apps data.

Functionality & Design

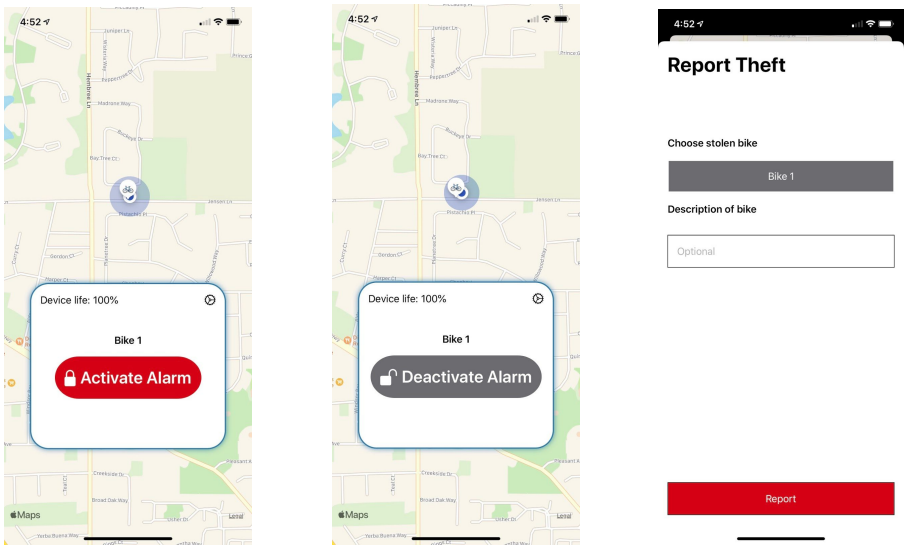
For all images of the mobile application, the iOS version of the application is used.

Hardware Device

Figure 1: Fully put together device

Device put together including speaker, Arduino Nano BLE Sense board, battery, and charging wire. Device is wrapped in shrink wrap. Reset button to restart the program on the board is located near the speaker, shown at the bottom of the device in Figure 1.

App Design



Figures 2, 3, 4: Activating the alarm (left), deactivating the alarm(middle), reporting a stolen bike(right)

Arming and Disarming Device

The button that users can use to enable or disable the alarm seen in figures 2 & 3.

Location of Bike

Where the user was last connected to a bike with relation to current location on the map seen in figures 2 & 3. The user's location is shown with the blue with the dot on the map and the white dot with the bike logo represents the location where the bike was last connected to the mobile phone.

Device Battery Life

Shows battery life which is calculated based on use of the device. Shown above the arming button in the top left corner of the settings box as seen in figures 2 & 3.

To choose some of the below functionality, tap the settings button in the top right of the box in figures 2 & 3:

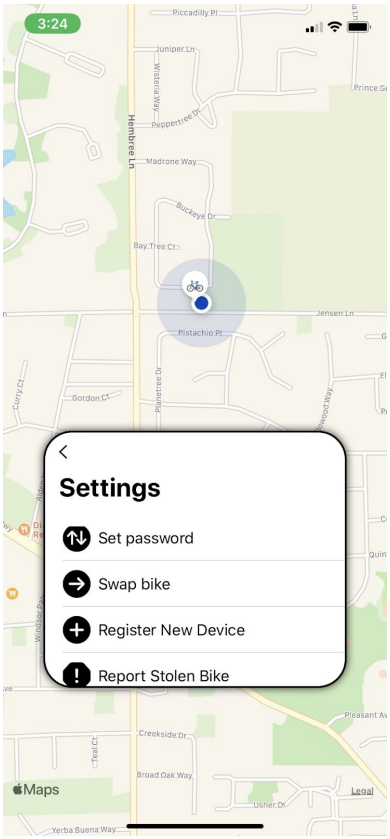


Figure 5: settings view for more functionality of the app

Configuring Motion Password

Clicking this button in figure 5 has the app tell the alarm to set the motion password of the device.

Choosing a Bike

Choosing a registered bike to be in use by pressing the button in figure 5 displays a list of all the user’s bikes registered with this application and can also switch which bike is active.

Registering New Device

Register a new bike by qr code or manually entering UUID can be done by pressing the button in figure 5. Pressing the button opens a camera that can be used to scan a QR code as well as an option to input the UUID manually.

Reporting Stolen Bike

Reports a registered bike of choice to a firebase database and adds description of stolen bike in the app view seen in Figure 4 after pressing the button in figure 5.

Troubleshooting

The hardware is complicated and is usually the cause of any malfunctions. When troubleshooting a user should start by resetting the device. The device can be reset by pressing the button on the device which is near the speaker.

FAQ

Where do I attach this device to my bike?
This device is designed to go in the frame underneath any bike's seat. Simply remove the bike seat and slide the device into the frame and then reattach the bike seat.

How do I charge the battery of the device?
There is a cord at the end of the device that can be plugged into with a micro-usb cable. By doing this it will charge the device.

How do I connect the device to my phone?
The app can be downloaded from the Google play store or from the App store. Once downloaded you must scan the devices qr code or manually enter the 4 digit id in order to register the device to your phone. Once registered the app will connect to the device as long as it is the selected device.

Contact Information

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See appendix for Github Repositories.

Appendix: Design Document

Technology Survey

Native vs. Cross platform
Native Pros:

- Better suited for working with bluetooth
- We have knowledge of the platforms
- Better established community and documentation

Native Cons:

- Develop app twice

Cross Platform Pros:

- Develop app once
- Expand skill set

Cross Platform Cons:

- Learning curve for new technologies

System Architecture Overview

To achieve the end goals of our product, we used a low energy bluetooth chip consisting of the nrf52840 core. This BLE chip is on the Arduino Nano 33 BLE Sense which has multiple sensors such as the LSM9DS1. To program the chip, we used the Arduino IDE and its libraries to develop the hardware side. Alongside this, we also used mobile application development from both the iOS and Android development kits in order to implement the application.

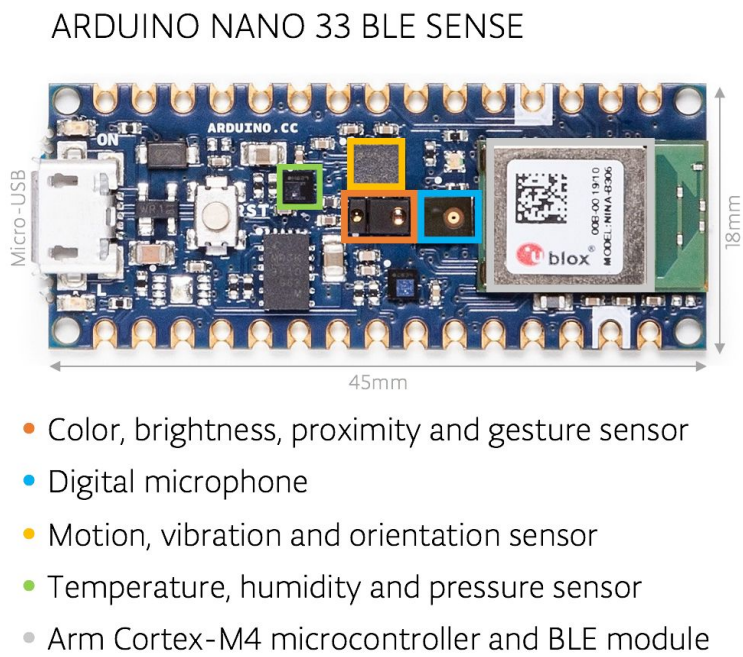

Figure 6: Arduino Nano 33 BLE Sense chip

High Level Requirements

- Bluetooth alarm activated when user is extends past a certain radius from bike
- Bluetooth alarm is activated when a bike is in motion without being in proximity with the user. Gives a little beep even when there is very little motion (i.e. someone nudges the bike)
- Need the battery to be small, rechargeable, able to last significant amounts of time
- Have a combination that the owner can do to prevent the alarm from going off (when phone dies)
- App requirements
    - Bluetooth pairing
    - Turn the alarm system on and off from the app
    - GPS location for where bike is/last known location when paired
- Potential for use with multiple bikes owned by user
- Potentially have a very low cost web backend to keep track of registered devices
- Transferable pairings of bike and app to friend
- Two factor authentication
- One time bike unlock combo

User Stories

- As a user I want to be able to pair my device with my phone so that I can track my bike and only my bike
- As a user I want an alarm to go off when my bike is in motion when I'm not nearby to deter anybody else from riding my bike
- As a user, I want to be able to replace or recharge the batteries on my alarm system with ease.
- As a user I want to be able to hide the device inside or on my bike so that it cannot easily be removed
- As a user I want to not worry about my device dying so that the device is easy to use
- As a user I want to know when my device has died or is out of battery
- As a user I want to know where I last had my bike so that I can find it if I lose it, or to verify that it is missing
- As a user, I would like to re-pair my device if I get a new phone
- As a user, I would like to temporarily disarm my bike so that my friend can ride it without the alarm going off
- As a user, I would like to be able to disable the alarm without the phone if my phone dies so that I can still ride my bike
- As a user, I would like my bike to not activate the alarm when somebody is not trying to steal the bike to false positive alarms or unnecessary disturbance

Github

https://github.com/rajsbhatia/Bluetooth-Anti-Bike-Theft-System
This repository also contains links to the repositories used for the Android and iOS application repositories.

Technologies Employed

- App
    - iOS
        - Xcode
        - Swift
        - UIKit
        - Core Bluetooth
    - Android
        - Android Studio
        - Java
- Arduino Nano 33 BLE Sense
    - Arduino IDE
        - C++
    - Accelerometer/Gyroscope
    - Speaker

Cost Analysis

- Developer account to deploy iOS app to app store: $99
- Arduino Nano 33 BLE Sense $25

Social/Legal Aspect

- Crowdsourcing of data for heat maps and data regarding common places people would steal bikes, store bikes, etc. would be saved after consent from the user
- If we use the feature of determining where a bike last was, that data will be stored with the client. A history of where the bike has been parked will not be saved.
- If the alarm system detects false positives frequently, the alarm system can be a disturbance
- Long process to get bluetooth device officially approved as an Apple certified device

Moving Forward

As bike theft is a continual problem in Davis, California and beyond, the project can be updated and refined to make the product more robust against issues that remain in the product and the mobile applications. Additionally, anyone can contribute to the project as it is an open-source project.

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